

3 (long) A cylindrical beaker of radius  $R$  and height  $H$  contains water up to height  $h_0$  when at rest, as shown in Fig. 3(a). When the beaker rotates about its axis at a constant angular velocity  $\omega$ , the water inside reaches a steady motion rotating at the same angular velocity as the beaker, as shown in Fig. 3(b). When rotating at an angular velocity  $\omega$ , the maximum water height above the bottom of the beaker is  $h_1$ , and the depth of water is large enough that the free surface never reaches the bottom of the beaker.

(a) By considering the radial pressure forces on a small fluid particle a distance  $r$  from the axis of rotation, or otherwise, derive a differential equation for the radial variations of the pressure. Integrate it to find the pressure difference between points A and B ( $p_B - p_A$ ) in Fig. 3(b). [8]

(b) By considering the hydrostatic variations of the pressure, obtain an expression for the height of the free surface above its lowest point,  $\Delta h(r)$ . [8]

(c) Using continuity, or otherwise, show that the maximum height of the liquid has the form

$$h_1 = h_0 + C_1 \frac{\omega^2 R^2}{g},$$

where  $g$  is the acceleration of gravity. Find the value of the constant  $C_1$ . What is the maximum possible  $\omega$  to avoid spilling water? [10]

(d) The conical beaker shown in Fig. 4 is tall enough that when rotating at large  $\omega$  the free surface does not reach the rim. However, this alone does not guarantee that no water will be spilled. What other condition does the curve  $\Delta h(r)$  need to satisfy at  $r = h_1$  for the analysis to hold, the solution to be stable, and no water to escape the volume? Express the condition in terms of  $h_1$ ,  $\omega$  and  $g$ . [4]

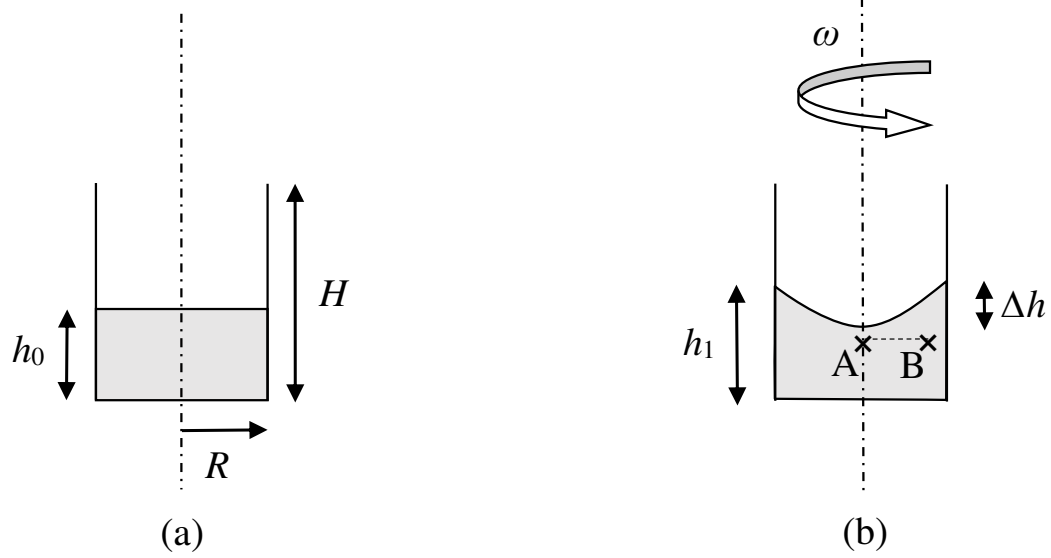


Fig. 3

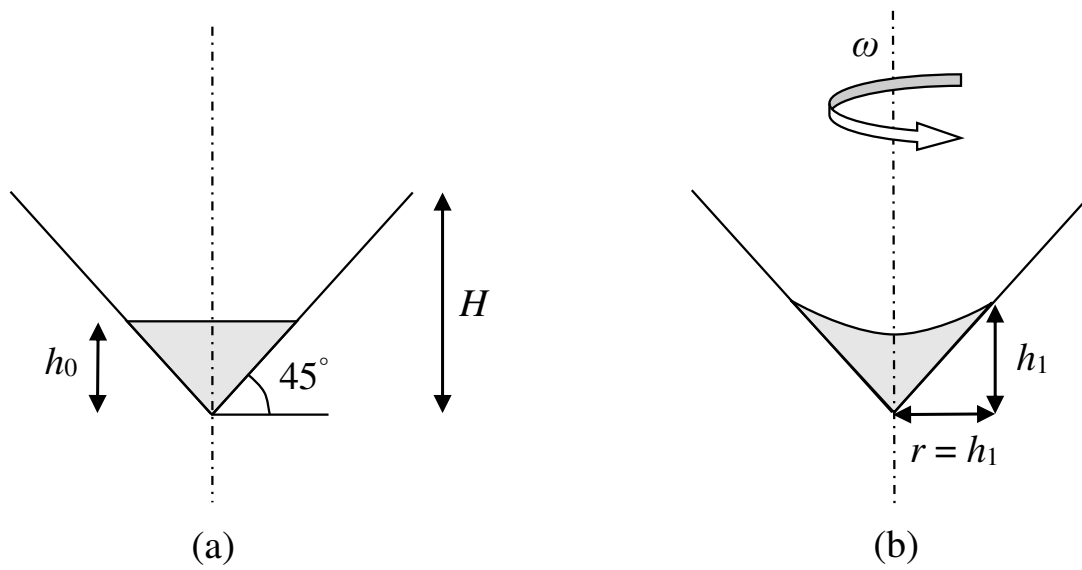


Fig. 4